Introduction

Phylogenetic trees depict the evolutionary relationships among sequences (species, strains, cells). They are central to taxonomy, molecular epidemiology, and cancer-clone tracing. `ape` is the standard R package for tree construction and manipulation, and `phangorn` adds maximum-likelihood and parsimony methods.

Prerequisites

Multiple sequence alignment; nucleotide or amino-acid substitution models.

Theory

Distance-based methods (neighbour-joining, UPGMA) compute pairwise distances from the alignment and build a tree that approximates those distances; fast but statistically crude.

Maximum-likelihood (ML) methods posit a substitution model (JC69, K80, GTR + Gamma + I) and find the tree topology + branch lengths maximising the likelihood of the observed alignment.

Bootstrapping resamples alignment columns to assess topology confidence; bootstrap values ≥ 70 % are commonly considered well-supported.

Assumptions

Alignment columns are homologous and independent; the substitution model captures key features (transition/transversion bias, rate heterogeneity).

R Implementation

```
library(ape)

# Load an example alignment shipped with ape
data(woodmouse)          # DNAbin alignment of 15 sequences, 965 sites

# Distance matrix under K80 model
d <- dist.dna(woodmouse, model = "K80")

# Neighbour-joining tree
nj_tree <- nj(d)
plot(nj_tree, main = "Neighbour-joining tree (woodmouse)",
     cex = 0.6, font = 1)
add.scale.bar()

# Bootstrap support
bp <- boot.phylo(nj_tree, woodmouse,
                 function(x) nj(dist.dna(x, model = "K80")),
                 B = 100, quiet = TRUE)
plot(nj_tree, main = "NJ tree with bootstrap support",
     cex = 0.6, font = 1)
nodelabels(bp, cex = 0.5, frame = "none", adj = c(1.3, -0.3))
```

Output & Results

A neighbour-joining tree with branch lengths in substitutions per site; bootstrap values annotated at internal nodes.

Interpretation

"The neighbour-joining tree of the cytochrome b sequences recovers the regional subclades with bootstrap support > 80 %, consistent with the geographic sampling structure."

Practical Tips

- For genome-scale data, use ML tools like `RAxML-NG` or `IQ-TREE` externally and import trees with `ape::read.tree()`.
- NJ is fast but sensitive to model misspecification; ML is the gold standard for small-moderate alignments.
- Use `phangorn::modelTest()` to pick a substitution model by AIC/BIC before running ML.
- Annotate trees with `ggtree` for publication-ready figures including metadata overlays.
- For large strain trees (SARS-CoV-2), use Nextstrain / `treetime` rather than base ape.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat